Project Proposal Guide (Weeks 7–8)

This worksheet guides the team in developing the Capstone/Software Engineering Project Proposal. It integrates the refined problem statement, findings from the Review of Related Literature (RRL), and the proposed solution concept.

Team Information

Project Title: DASIGConnect

Project Short Description (<20 words): A centralized platform for coordinating, validating, and scheduling DASIG social media content across multiple member institutions.

Team Code: 2526-sem2-it332-38

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PART 1: Introduction (Approx. 300 words)

The management of organizational social media presence has become a critical operational function in academic and research institutions. Effective and timely communication through social media platforms directly influences stakeholder awareness, institutional visibility, and community engagement (Kaplan & Haenlein, 2010). For multi-organizational networks such as the DOST Acadême–Science and Innovation Group (DASIG), which consolidates participation from multiple Higher Education Institutions (HEIs) including the Cebu Institute of Technology–University (CIT-U), Silliman University, and other member schools under DOST Region 7, the challenge of coordinating and publishing content across diverse contributors is particularly pronounced.

Preliminary observations of the DASIG Facebook page reveal recurring patterns of delayed event coverage, incomplete posts, and irregular publishing activity. Events held by member institutions are frequently reported several days after their occurrence, and some activities receive no coverage at all due to missing media assets or unclear submission responsibilities. This irregularity has resulted in reduced audience engagement and inadequate representation of the organizations' activities on the platform. The root cause of these issues can be traced not to a lack of content, but to the absence of a structured workflow for collecting, validating, and scheduling content from multiple independent sources.

Existing literature consistently demonstrates that structured content workflows and scheduling tools significantly improve the timeliness and consistency of organizational social media outputs (Turban et al., 2018). Studies on collaborative digital publishing further indicate that approval-based moderation mechanisms reduce errors and improve content quality in multi-contributor environments (O'Reilly, 2007). Additionally, research on social media management in academic organizations confirms that irregular posting patterns negatively affect institutional credibility and audience retention (Aral et al., 2013).

Recent advancements in artificial intelligence, particularly in image recognition and natural language generation, offer practical opportunities to further improve content workflow efficiency. AI-assisted caption generation based on uploaded images, along with intelligent content suggestions drawn from the media repository, can reduce the manual burden on contributors and improve the consistency of post quality without requiring significant technical expertise from end users (Radford et al., 2021; Li et al., 2022).

Despite these established findings, no documented system specifically addresses the multi-institution content coordination challenge faced by academic consortia such as DASIG, particularly one that integrates AI-assisted tools to support contributors throughout the content preparation process.

PART 2: Objectives

General Objectives

This project aims to achieve the following general objectives, structured by key system features, within the duration of two academic semesters:

1. **Foundation, Access Governance, and Content Authoring.** To increase content submission completeness to a ≥95% form completion rate on first submission attempt, through a secure, multi-organization content authoring and access-governance system supporting at least two (2) DASIG member institutions and three primary user roles—Contributor, Moderator, and Administrator—delivered by the end of the first development phase.
2. **Approval Workflow, Media Governance, and Communication.** To eliminate unapproved content publication and reduce submission-status notification delay, through a content approval, media asset management, and notification subsystem that (a) enforces a 100% Moderator-reviewed approval step before any Contributor submission is published, (b) reduces notification delivery time to within 5 minutes of state change in ≥95% of cases across in-app, email, and messaging channels, and (c) improves media asset accessibility and brand consistency through a searchable, tagged, and automatically watermarked library indexed by institution and event, achieving a System Usability Scale (SUS) score of ≥70 from at least 10 pilot users drawn from Contributors, Moderators, and Administrators by the end of the second development phase.
3. **Scheduling, Automated Publishing, and AI-Assisted Content Support.** To improve posting timeliness and reduce caption drafting effort, through an automated scheduling, publishing, urgent-event, and AI-assisted content subsystem that (a) increases scheduled post publish success rate to ≥95% within ±5 minutes of the assigned time, (b) supports expedited same-day publication for time-sensitive events, (c) achieves a ≥60% contributor acceptance or accept-with-edits rate on AI caption suggestions generated for ≥90% of uploaded images, and (d) attains an average AI suggestion quality rating of ≥4.0 / 5.0 across at least 20 evaluation submissions, by the end of the third development phase.

Specific Objectives

To achieve the general objectives, the project will accomplish the following feature-specific tasks, which serve as sub-objectives:

For General Objective 1: Foundation, Access Governance, and Content Authoring

1.1. To enable multiple designated personnel to jointly administer the platform while preserving a single point of accountability for account removal, by implementing an administrator provisioning mechanism in which any administrator may invite new administrators, but only one designated role-holder may revoke administrator access, with a formal, auditable mechanism to transfer that authority when needed.

1.2. To standardize institutional participation in the network and support content issued on the network's behalf rather than a specific member school, by implementing an institution registration and lifecycle management mechanism covering creation, detail editing, deactivation, and reactivation, including a protected default institution reserved for network-wide announcements.

1.3. To enable independent content submission from at least three distinct DASIG member institutions without coordinator intervention, by implementing a secure, institution-scoped contributor provisioning mechanism that invites, deactivates, and reactivates contributor accounts on an administrator's behalf.

1.4. To ensure institution-level data isolation and eliminate unauthorized cross-institution access, by implementing a secure authentication and session-management mechanism that provisions isolated workspaces per institution, with role-based routing to either the institutional workspace or the network admin console upon login.

1.5. To reduce contributor drafting effort and improve caption consistency, by providing a content authoring interface with reusable post templates covering common DASIG content types (event announcements, recaps, calls for participation, and institutional spotlights) that pre-fill structural elements of the caption.

1.6. To reduce manual caption-writing burden and support brand-consistent stylistic presentation, by integrating an AI-assisted caption generation tool, optionally guided by contributor instructions, alongside a text-styling utility for visual emphasis in captions.

1.7. Develop a media attachment interface with server-enforced validation requiring at least one media upload (photo or video) and assignment of that media to an organizational grouping, supporting device upload, repository browsing, and AI-suggested relevant assets, achieving a verified $\geq$95% form completion rate on submitted records during pilot testing.

1.8. To increase scheduling efficiency and eliminate publication time conflicts across member institutions, by implementing a self-service scheduling mechanism, informed by peak-engagement time recommendations, that enables contributors to select a publication slot directly during submission, with built-in conflict detection.

1.9. To support urgent, same-day event coverage without compromising the approval requirement, and to finalize routine submissions into the approval pipeline, by implementing a submission-finalization mechanism offering both standard scheduled routing and an expedited pathway for time-sensitive events that publishes immediately upon approval.

1.10. To distribute routine review work without binding reviewers to a single institution, by implementing a network-wide Moderator role with secure provisioning, activation, deactivation, and reactivation controls, access to the approval queue and publication calendar, and no authority over Administrator governance, institution lifecycle management, or system configuration.

For General Objective 2: Approval Workflow, Media Governance, and Communication

2.1. To reduce contributor effort in organizing uploaded media and to preserve institutional visual assets for reuse, by implementing a direct media library upload mechanism with automatic and manual assignment of assets to event-based groupings.

2.2. To improve media asset retrieval speed to $\leq$2 seconds, by implementing a semantic search and filtering mechanism supporting discovery by tag, event context, and visual similarity across libraries of up to 1,000 assets.

2.3. To increase asset reuse across submissions and preserve institutional accountability for media lifecycle actions, by implementing reuse-history tracking per asset and lifecycle safeguards governing asset deletion.

2.4. Implement a Moderator approval interface that eliminates unapproved or erroneous Contributor posts by enforcing a 100% Moderator-reviewed approval step before publication. Moderators may edit minor content details while a submission remains in review, but editing does not approve or otherwise finalize it. The terminal review actions are Approve, Request Revision with remarks, and Reject.

2.5. To maintain consistent visual branding across all published content without added manual effort, by implementing an automated watermarking mechanism that applies a configurable network-wide or institution-specific brand overlay to approved photo content at the point of publication, adaptable to varying image dimensions.

For General Objective 3: Scheduling, Automated Publishing, and AI-Assisted Content Support

3.1. To provide consolidated visibility into network-wide publishing activity while preserving institutional content confidentiality, by implementing a master scheduling calendar with role-differentiated visibility of submission timing and content detail.

3.2. Integrate the Facebook Graph API using Standard Access to automatically publish approved and scheduled content to the connected Facebook Page, achieving a publish success rate of $\geq$95% within ±5 minutes of the scheduled time, with a manual publishing recovery pathway for content that fails automated publication.

3.3. To reduce response time and minimize workflow delays, by implementing a role-aware, multi-channel notification system that alerts Contributors, Moderators, and Administrators of relevant submission status changes, review assignments, publishing outcomes, and platform events within 5 minutes in $\geq$95% of cases across in-app, email, and direct messaging channels as applicable.

3.4. Create a role-scoped social engagement analytics dashboard that improves operational visibility by tracking posting frequency, submission-to-publish duration, content completeness rate, review turnaround, and AI feature performance. Contributors view institution-scoped information, while Moderators and Administrators view network-wide operational information; the system targets a minimum of 4 posts per month per institution and a completeness rate of at least 95%.

3.5. To support proactive platform maintenance and minimize service disruption, by implementing a system health monitoring interface tracking third-party integration status, background process reliability, and infrastructure resource usage.

3.6. To preserve institutional accountability for all governance-relevant actions, by implementing an immutable, filterable audit trail covering account management, content approval decisions, content edits made during approval, and configuration changes.

Research Questions

The following research questions will guide the evaluation of the proposed system:

- To what extent does the implementation of DASIGConnect reduce the average posting delay (days from event to publication), compared to the pre-implementation baseline, during the pilot deployment period?
- By what percentage does the structured submission and approval workflow increase the content completeness rate (percentage of posts published with all required elements), compared to the baseline?
- What SUS score does DASIGConnect achieve among DASIG Contributors, Moderators, and Administrators following the pilot deployment period, and does this meet the target threshold of 70 or above?
- During the pilot deployment period, does DASIGConnect increase posting frequency and short-term consistency on the DASIG Facebook page compared to an equivalent pre-implementation period, and what does the observed trend suggest about its potential longer-term impact?
- To what degree does the AI-assisted caption generation feature reduce caption drafting time, and how do pilot contributors rate the relevance and quality of AI-generated suggestions during the deployment window?
- To what extent does the expedited submission pathway for time-sensitive events function as intended during the pilot deployment — enabling same-day approval and publication when used — and what does observed usage (frequency, approval turnaround) suggest about its value for future full-scale deployment?
- How consistently does the automated watermarking feature apply correct, properly-scaled branding across approved photo content during the pilot deployment, and what administrator feedback emerges regarding its effect on manual branding effort?

PART 3: Methods

Proposed Solution Concept

The proposed solution is a web-based Social Media Content Workflow and Scheduling Management System designed specifically for the DASIG multi-institutional network. The system addresses the identified problem by replacing the current informal and ad hoc coordination approach with a structured workflow built around three primary roles. Contributors from member institutions—including faculty, student organization

officers, and institutional communications officers—prepare event content, attach media, and select a publication time within system-enforced scheduling rules or use an expedited pathway for time-sensitive events. Network-wide Moderators review Contributor submissions through the approval queue, may correct minor content details while keeping the submission in review, and then approve, request revision, or reject the submission. Administrators manage users and institutions, oversee analytics and system configuration, monitor publication outcomes, and resolve operational exceptions such as failed publications; routine content approval is not their primary workflow. The system will also maintain a centralized, semantically searchable media repository with automated event-based organization and brand-consistent watermarking, ensuring that assets are preserved, discoverable, and reusable across submission cycles.

To further support contributors during the content preparation stage, the system will incorporate AI-assisted features in a targeted and non-intrusive manner. When a contributor uploads an image during submission, the system will use a vision-capable language model to analyze the image and generate suggested captions relevant to its content, optionally guided by contributor instructions, which the contributor may accept, modify, or discard. Additionally, the system will offer an intelligent media recommendation panel that suggests existing photos from the repository based on the event context entered, helping contributors identify supplementary media without manually browsing the full library. These AI features are designed to reduce friction in the submission process rather than replace contributor judgment.

Development Methodology

The project will adopt the Agile Software Development methodology, specifically utilizing iterative sprint cycles guided by the Scrum framework. This approach is appropriate for the project given the need for continuous stakeholder feedback from multiple member institutions and the iterative refinement of features based on real-world usage patterns. Development will proceed in two-week sprints, with each sprint producing a testable increment of the system. Sprint reviews will involve designated DASIG stakeholders to validate progress and realign priorities. This methodology supports structured development, promotes early detection of functional gaps, and allows the team to adapt to changing requirements without disrupting the overall project timeline.

Validation Approach

The system will be evaluated using a combination of functional testing, controlled performance measurement, role-based user acceptance testing (UAT), and pilot-deployment analysis. The evaluation will be guided primarily by ISO 9241-11, assessing system usability in terms of effectiveness, efficiency, and satisfaction within the defined context of use. Effectiveness will be measured through task-completion outcomes, efficiency through Time-on-Task and relevant system-response measurements, and satisfaction through post-task ratings and the System Usability Scale (SUS).

Role-specific validation scenarios will be conducted for Contributors, Moderators, and Administrators based on their respective responsibilities within DASIGConnect. Contributor tasks will evaluate content submission, media attachment and retrieval, scheduling, submission-status awareness, and applicable AI-assisted features. Moderator tasks will evaluate approval-queue navigation, submission review, non-terminal editing, terminal review actions, and publication-calendar interpretation. Administrator tasks will evaluate institution and account management, publication-status monitoring,

failed-publication recovery, and operational oversight. Task outcomes will distinguish successful independent completion, successful completion with assistance, unsuccessful completion, and tasks that could not be performed because the required condition was unavailable.

Functional and technical requirements that cannot be validly measured through questionnaires will be evaluated using controlled tests, system logs, application timestamps, and audit records. These include role-based access control and institution isolation, mandatory submission validation, media-search performance, scheduling conflict prevention, Moderator approval enforcement, notification-delivery time, watermark consistency, automated publishing reliability, audit-trail integrity, and system-health monitoring. Questionnaire responses will therefore provide user-experience evidence but will not substitute for technical evidence required to validate performance, security, governance, or operational targets.

Performance measurement will compare pre-implementation baseline data gathered through retrospective observation of the DASIG Facebook page against post-implementation metrics collected during the pilot deployment period. Primary operational indicators will include average posting delay, content completeness rate, posting frequency, submission-to-publish duration, review turnaround time, and other applicable system-generated measures. Posting frequency will be interpreted according to the duration of the pilot and normalized where necessary when comparing it with an equivalent pre-implementation period.

User efficiency will additionally be evaluated through Time-on-Task measurements for representative role-specific workflows and media-search duration where applicable. Because the pilot population is expected to be relatively small, medians, ranges, and other descriptive measures will be emphasized, while system timestamps and recordings will be used to corroborate self-reported task durations when available.

AI-assisted feature effectiveness will be evaluated using both system-generated evidence and participant feedback. Caption adoption will be measured from valid AI-caption interactions by determining the proportion of generated suggestions accepted unchanged or accepted after editing. Caption quality and media-recommendation relevance will be evaluated only among participants or submissions that actually encountered the corresponding AI feature. Perceived usefulness will be assessed through structured items addressing whether AI assistance improved task speed, output quality, relevance, and willingness to use the feature in future workflows. AI generation availability, coverage, and response time will be measured through system logs or controlled testing rather than participant recall.

User acceptance will be measured using the ten-item System Usability Scale across Contributors, Moderators, and Administrators, targeting an aggregate SUS score of at least 70 from at least 10 valid pilot participants. SUS scores will be calculated using the standard scoring procedure, with aggregate results reported alongside role-specific results and valid sample sizes where appropriate. Supplementary post-task satisfaction measures will assess perceived ease, confidence, usefulness, and overall satisfaction.

The expedited submission pathway will be evaluated descriptively based on observed usage during the pilot deployment period, including frequency of use and approval turnaround time when the pathway is exercised, rather than against a fixed numerical target because the number of time-sensitive events cannot be predicted in advance. Automated

watermarking will likewise be evaluated through consistency checks across approved photo content and Administrator feedback regarding its effect on manual branding effort.

Qualitative feedback from open-ended questionnaire items, task observations, and participant comments will be coded into recurring themes such as navigation, scheduling, media management, review actions, status visibility, AI quality, and system performance. Significant issues will be triangulated with available screen recordings, application logs, and system records to identify their likely causes and determine their impact on the corresponding MVP objectives.

For the pre-implementation baseline, content completeness will be assessed using externally observable Facebook post criteria, specifically the presence of at least one photo and a caption of at least 50 characters. Post-implementation completeness will additionally account for server-enforced requirements, including event date and mandatory media assignment to an organizational grouping. This difference in available evidence will be explicitly acknowledged when comparing baseline and post-implementation completeness rates.

PART 4: Expected System

Minimum Viable Product (MVP) Features

The MVP of the DASIGConnect Content Workflow and Scheduling Management System will include the following core functionalities:

- Three-role, multi-organization access control distinguishing institution-scoped Contributors, network-wide Moderators, and Administrators, including secure provisioning and lifecycle management for each role and a single designated Administrator with sole authority over Administrator account removal and a formal succession mechanism to transfer that authority.
- Institution registration and lifecycle management, including a protected default institution for content issued on the network's behalf.
- Content submission form with fields for event title, date, caption, and media upload (photos and videos), supported by reusable content templates.
- AI-assisted caption suggestion panel that analyzes uploaded images and generates contextually relevant caption drafts, optionally guided by contributor instructions, for review and editing.
- Media attachment interface supporting device upload, repository browsing, and AI-suggested relevant assets, with mandatory assignment to an event-based organizational grouping.
- Self-service scheduling with peak-engagement time recommendations and network-wide conflict detection, plus an expedited pathway for urgent, same-day events.
- Moderator approval interface allowing network-wide Moderators to inspect Contributor submissions, edit minor content details without finalizing the review, and complete one of three terminal actions: approve, request revision with remarks, or reject.
- Centralized, semantically searchable media library organizing uploaded files by institution and event, with automated and manual grouping, and reuse-history tracking.
- Automated brand watermarking applied to approved photo content.
- Master scheduling calendar with role-differentiated visibility across institutions.
- Automated publishing to the connected Facebook Page, with a manual recovery pathway for failed publications.

- Multi-channel notification system (in-app, email, and direct messaging) for submission and publishing status changes.
- Analytics dashboard displaying posting frequency, pending submissions, average submission-to-publish time, and content completeness.
- System health monitoring for third-party integrations and background processes.
- Immutable audit trail of governance-relevant actions.

High-Level System Workflow

The high-level workflow of the system proceeds as follows. A Contributor from a DASIG member institution logs into the system and creates a content submission by attaching media assets and completing the required event details. Upon request, the AI caption module analyzes uploaded images and presents suggestions that the Contributor may accept, edit, or discard. The system may also recommend related repository images based on event context, and attached media is organized into an event-based grouping. The Contributor selects a publication time informed by peak-engagement recommendations and validated against network-wide scheduling rules, or chooses the expedited pathway for a time-sensitive event. The completed submission enters the network-wide Moderator approval queue. A Moderator verifies completeness and accuracy, may save minor edits while keeping the item in review, and then approves, requests revision with remarks, or rejects the submission. Upon Moderator approval, the system publishes the content at the designated time—or immediately for an expedited submission—and applies the configured watermark to approved photo content. Administrators monitor publication outcomes, manage users and institutions, review analytics and system health, maintain configuration, and resolve failed-publication exceptions. The analytics dashboard continuously reflects submission status, review turnaround, posting frequency, and content completeness according to each role's permitted scope.

PART 5: Discussion

Scope

The system is scoped to support the social media content coordination operations of the DASIG network under DOST Region 7, specifically for its official Facebook page. It supports three primary application roles: institution-scoped Contributors who prepare and submit content; network-wide Moderators who review the standard approval queue; and Administrators who manage users, institutions, analytics, configuration, system health, publication monitoring, and operational exceptions. The system will accommodate content submission from a defined set of member institutions identified by DASIG, plus a protected default institution for network-wide announcements. Supported content types include event announcements, event highlights, and activity recaps, with media support for image and video uploads. The scheduling module covers self-service Facebook post scheduling within the platform, including an expedited pathway for time-sensitive events. The analytics dashboard will cover a defined set of KPIs relevant to posting timeliness, review performance, consistency, and platform operational health. The AI-assisted features are limited to image-based caption suggestion using a pre-integrated language model API and semantic media recommendation based on vector similarity between event context and repository content; the system does not include autonomous content publishing decisions, AI-generated images, or content moderation through automated means. The system does not

include integration with social media platforms other than Facebook in its initial version. Facebook API integration will be configured for Standard Access under Meta's developer platform, which supports all required publishing operations for users with administrative roles on the connected Facebook Page. Full public post visibility is contingent on the completion of Meta Business Verification by the DASIG organization and the subsequent transition of the application to Live mode.

Limitations

The project is subject to the following constraints. First, the system requires reliable internet connectivity at Contributor, Moderator, and Administrator endpoints, which may affect usability in areas with poor network infrastructure. Second, the evaluation period for post-implementation KPI measurement is limited to the available deployment window within the second semester, which may not capture long-term usage patterns. Third, the baseline data for pre-implementation metrics will be gathered through retrospective observation of the DASIG Facebook page posting history, which may not fully reflect all coordination activities. Fourth, the automated publishing functionality of DASIGConnect depends on the Facebook Graph API operating under specific access and visibility conditions that are subject to Meta's platform policies. During the development and evaluation phases of this project, the system's Facebook application will operate under Development mode as defined by Meta's developer platform—a configuration that restricts the visibility of API-published posts to users who hold assigned roles on the registered Meta Developer Application, specifically Page administrators and designated application testers. This means that during the project's deployment and evaluation window, posts published through DASIGConnect's scheduling module will be visible in the DASIG Facebook Page feed only to authorized users with developer or administrator roles, and not to the general public or page followers. This behavior is a documented and expected characteristic of Meta's developer platform, not a functional defect of the system. To transition to full public visibility—wherein all published posts appear on the DASIG Facebook Page's public feed accessible to any Facebook user—the DASIGConnect application must be switched from Development mode to Live mode on the Meta Developer Platform. This transition requires Meta Business Verification, a formal process in which the operating organization—in this case, the DOST Acadême–Science and Innovation Group (DASIG) under DOST Region 7—must submit official business credentials, documentation, and contact information to Meta for identity verification. This verification process is governed entirely by Meta's policies and is contingent on DASIG's organizational decision to formally adopt and deploy DASIGConnect as an operational tool. As this decision falls outside the jurisdiction and timeline of the academic capstone project, the transition to Live mode and full public post visibility is treated as a post-deployment requirement dependent on DASIG's institutional authorization. The proposed system is designed such that the Development-to-Live transition will require no changes to the codebase—only the completion of the organizational verification process on Meta's platform. Furthermore, once DASIG grants the system access to the official DASIG Facebook Page as an administrator-level integration, the Page Access Token configuration will be updated to point to the production page without requiring any architectural modifications to the platform. Fifth, the accuracy of AI-generated caption suggestions is dependent on the capabilities of the integrated language model API and may vary based on image quality and content clarity; Contributors retain full editorial control over all submitted captions. Sixth, given the limited duration of the pilot deployment period, evaluation of features with inherently lower or less predictable usage volume during this period—specifically the expedited submission pathway and automated watermarking—will rely on descriptive reporting of observed usage and feedback from the role responsible for the affected workflow rather than statistically powered outcome targets.

Expected Contribution

The proposed system is expected to contribute to the DASIG network's operations by replacing the current informal, fragmented coordination approach with a structured, traceable, and efficient digital workflow. The primary contribution is the measurable reduction in posting delays and improvement in content completeness, directly addressing the identified operational problem. The three-role model separates content preparation, review, and governance responsibilities: Contributors prepare institution-scoped content, Moderators provide network-wide editorial review, and Administrators manage organizational access, operational oversight, analytics, and exceptions. The integration of AI-assisted caption generation, semantic media recommendation, and automated brand watermarking further reduces manual effort while preserving human editorial oversight. The system's governance mechanisms—role-based access control, auditable Administrator succession, and an immutable action log—provide a model for institutional accountability applicable beyond DASIG's specific context. Beyond operational improvement, the system contributes a replicable workflow model applicable to other multi-institutional academic networks facing similar social media coordination challenges. From an academic standpoint, this project addresses a documented gap in existing literature by producing a purpose-built solution for multi-organization content coordination in an academic consortium context, supporting future research on AI-augmented digital workflow systems in similar organizational structures.

PART 6: Traceability Matrix

RRL Finding / Theme	Identified Gap	Research Question	Proposed Function
Social media content scheduling improves posting consistency and audience engagement (Kaplan & Haenlein, 2010)	Lack of structured, self-service scheduling in multi-contributor organizations	RQ1, RQ4	Self-service scheduling with peak-time recommendations and conflict detection
Centralized digital workflow systems reduce coordination errors in multi-stakeholder environments (Turban et al., 2018)	No centralized submission and approval process across member institutions	RQ2	Multi-organization content authoring and institution-scoped access governance
Approval-based content moderation improves quality and reduces errors in collaborative publishing (O'Reilly, 2007)	No dedicated network-wide editorial review role or approval layer before content is published	RQ2	Moderator approval workflow with non-terminal editing and terminal approve, request-revision, and reject decisions

Inconsistent social media presence negatively affects organizational visibility and stakeholder trust (Aral et al., 2013)	DASIG page activity is irregular due to uncoordinated contributions	RQ4	Analytics dashboard with posting frequency and consistency tracking
Media asset management systems improve retrieval and reuse of organizational content (Baca, 2016)	Photos and media files are dispersed across contributors with no centralized repository	RQ2	Centralized, semantically searchable media library with automated event grouping
Large language models and vision-language models can generate contextually relevant text from images, reducing manual content creation effort (Brown et al., 2020; Radford et al., 2021)	Contributors lack support tools for drafting captions, leading to inconsistent post quality and delayed submissions	RQ5	AI caption suggestion module and intelligent media recommendation feature
Timely feedback mechanisms in digital workflow systems reduce bottlenecks and improve contributor responsiveness in multi-stakeholder environments (Turban et al., 2018)	No automated status notification system exists to alert contributors when submissions are reviewed, causing uncertainty and delayed follow-up actions	RQ1	Multi-channel (in-app, email, direct messaging) submission-status notification system
The System Usability Scale (SUS) provides a validated, technology-independent instrument for measuring perceived usability of interactive systems (Brooke, 1996)	No standardized usability evaluation method has been applied to assess whether the system interface is acceptable to diverse users across multiple DASIG member institutions	RQ3	SUS-based user acceptance evaluation methodology

Vision-language models can assign categorical labels to images with high accuracy, enabling automated organization of visual content at scale (Radford et al., 2021)	Uploaded media assets have no automated categorization, requiring manual tagging and making retrieval and reuse across submission cycles inefficient	RQ2	AI-powered image classification, auto-tagging, and album auto-matching
General-purpose social media management platforms (e.g., Meta Business Suite) provide single-organization publishing and scheduling tools, without native support for multi-organization submission routing, institution-level content isolation, or tiered approval workflows*	DASIG's existing alternative — Meta Business Suite — supports one-account publishing but has no mechanism for routing content from multiple independent institutional contributors through a structured, accountable approval chain before publication	RQ2, RQ4	Purpose-built multi-institution submission, approval, and attribution workflow (including network-wide default attribution and per-institution isolation), as a targeted alternative to generic single-organization tools
(no direct RRL support currently cited)	No mechanism exists for institutions to expedite coverage of live, time-sensitive events without bypassing the approval requirement entirely	RQ6	Expedited (fast-track) submission pathway, prioritized in the approval queue, publishing immediately upon approval
(no direct RRL support currently cited)	Manually applied branding is inconsistent and adds contributor workload; no automated mechanism ensures institutional visual identity across all published photo content	RQ7	Automated, configurable watermarking applied at the point of publication

PART 7: References

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